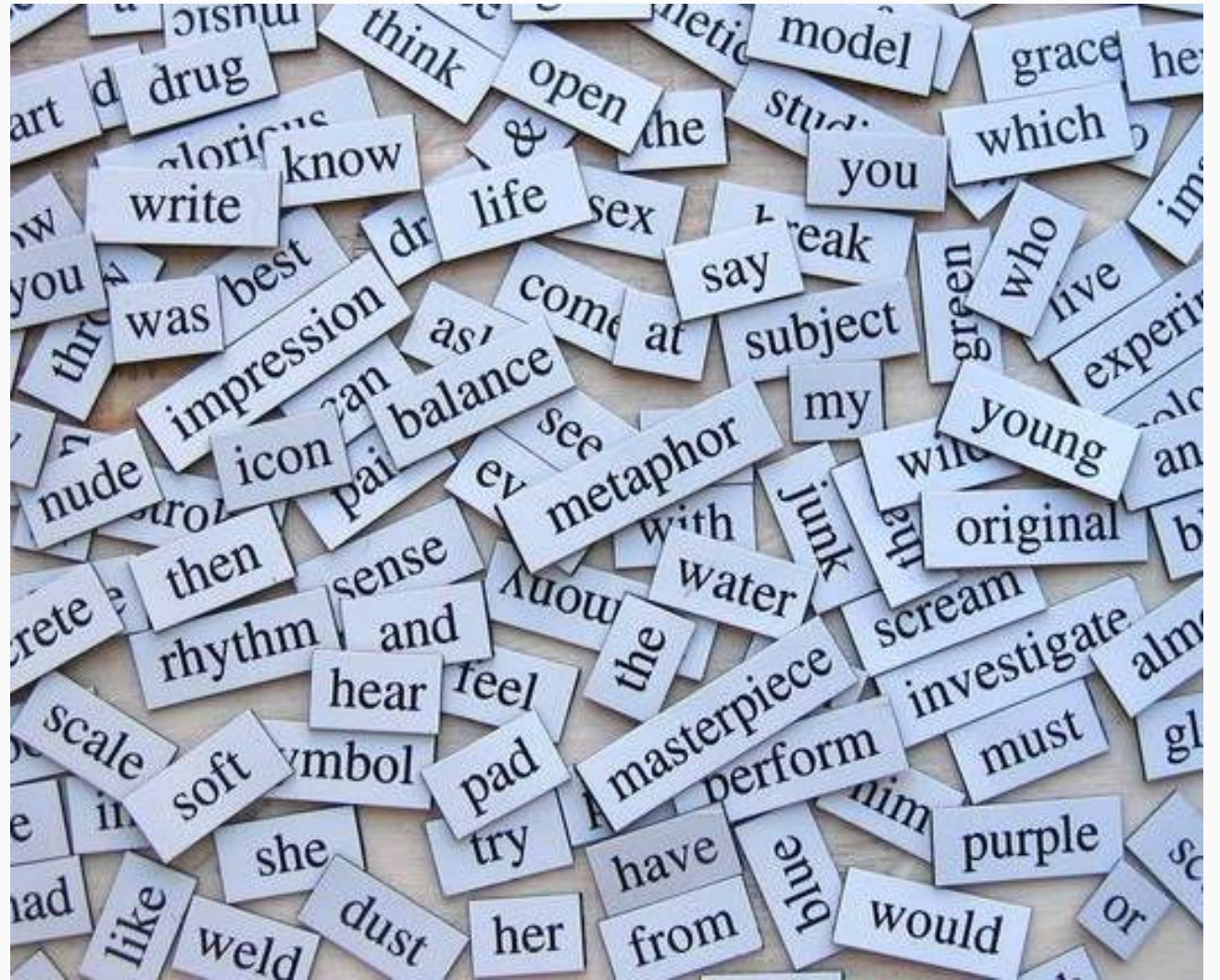


NATURAL LANGUAGE GENERATION



Contents

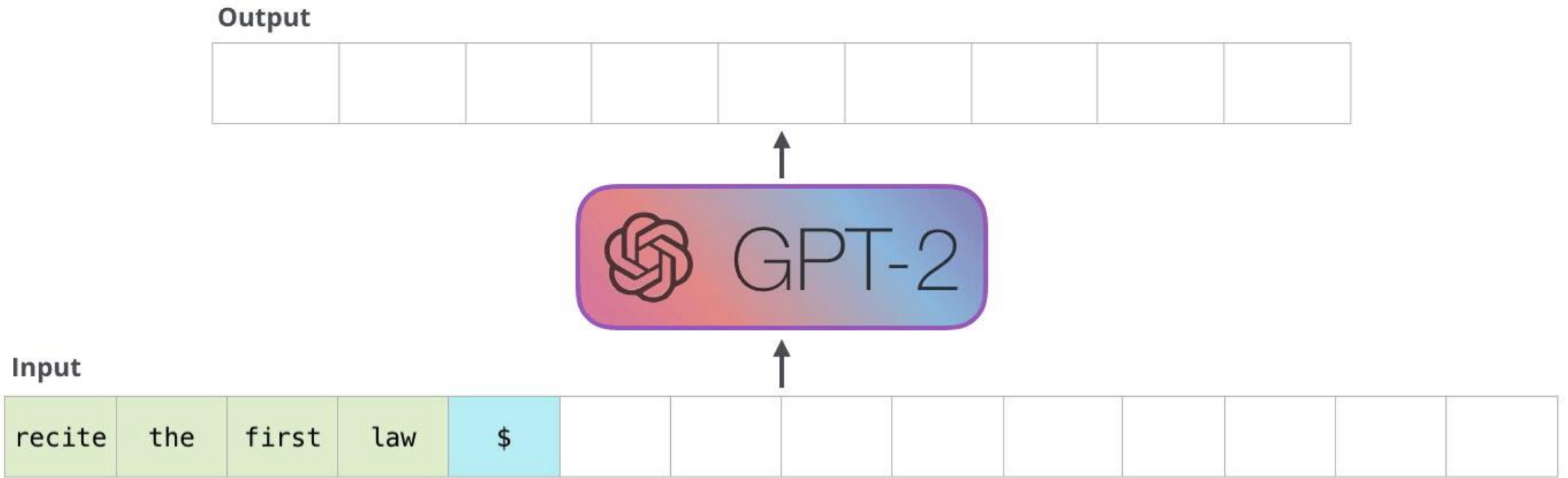
- **Brief Overview**
- **Decoding Algorithms**



BRIEF OVERVIEW

Generation Models

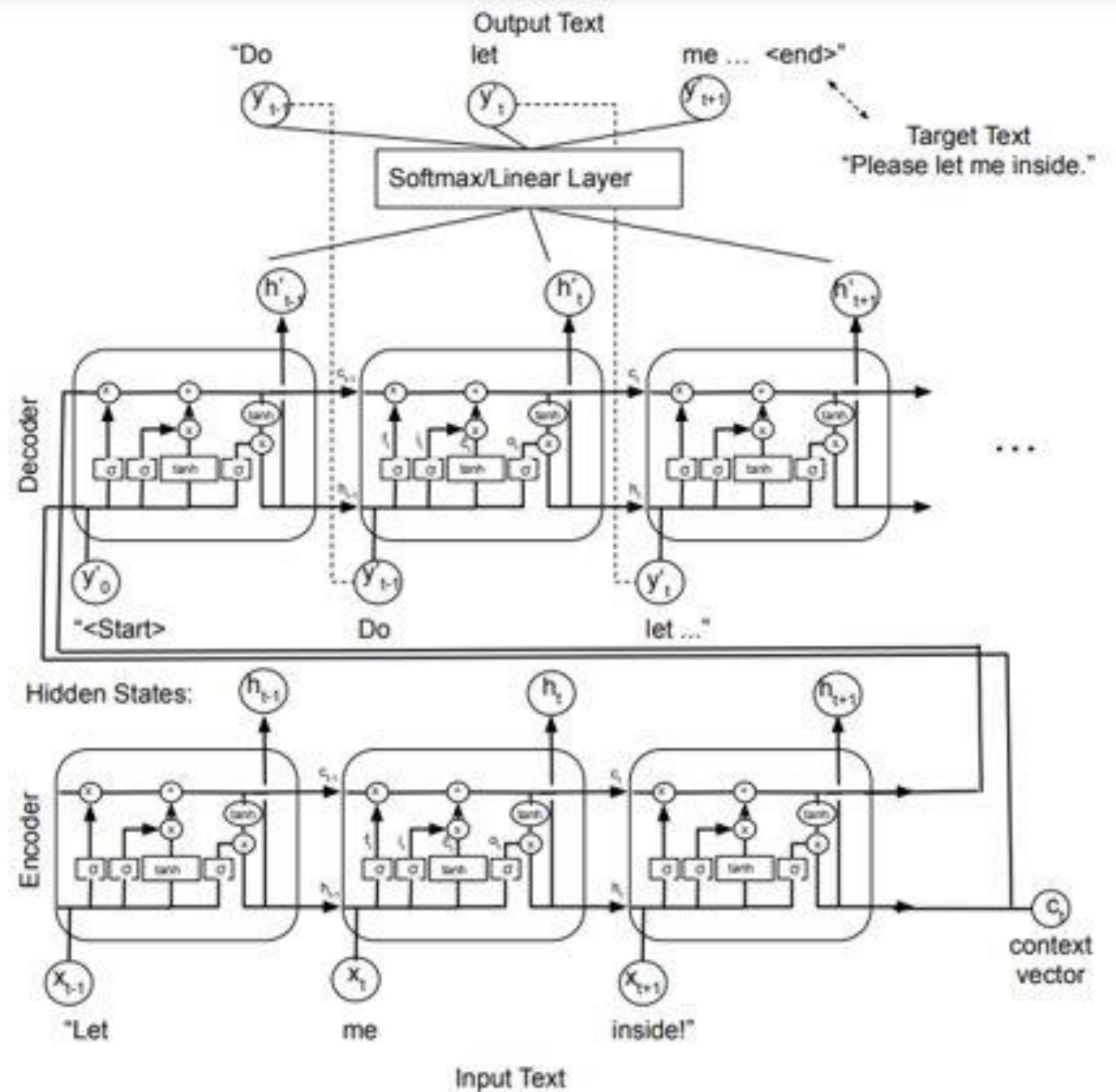
- Language Models
- RNNs, LSTMs
- Encoder-Decoder
- Transformer
- GPT, BART



AUTO-REGRESSION

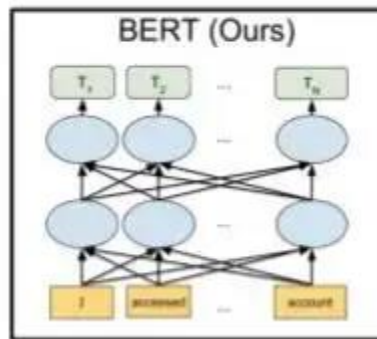
<https://jalammar.github.io/illustrated-gpt2/>

RNN, LSTM



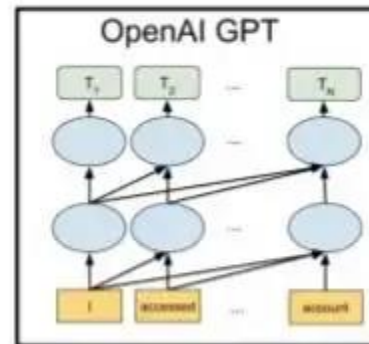
Encoder only

- BERT
- RoBERTa
- Reformer
- FlauBERT
- CamemBERT
- Electra*
- MobileBERT
- Longformer



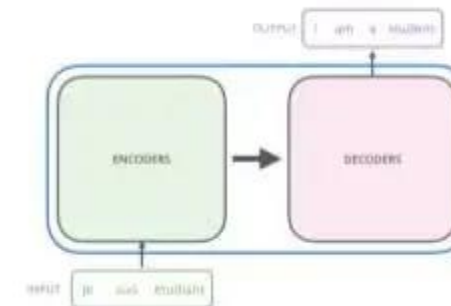
Decoder only

- Transformer-XL
- XLNet
- GPT series
- DialoGPT



Encoder + Decoder

- Transformer
- XLM
- T5
- BART
- XLM-RoBERTa
- Pegasus
- mBART

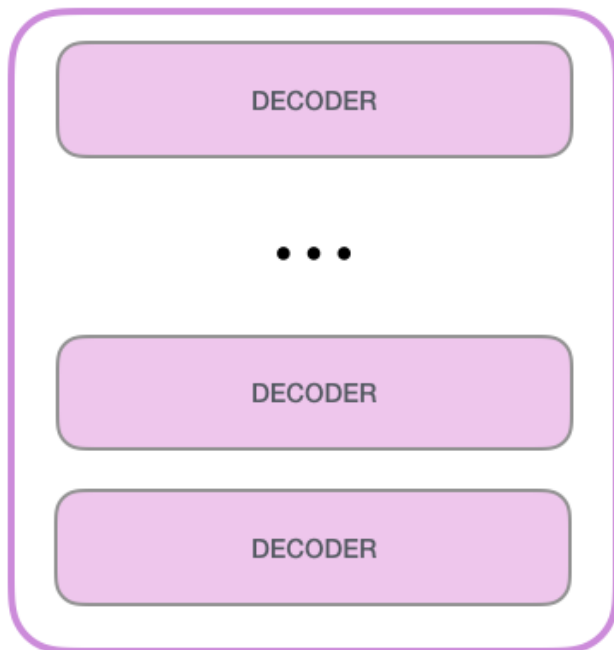


Illustrations are from: <https://ai.googleblog.com/2018/11/open-sourcing-bert-state-of-art-pre.html> and <http://jalammr.github.io/illustrated-transformer/>

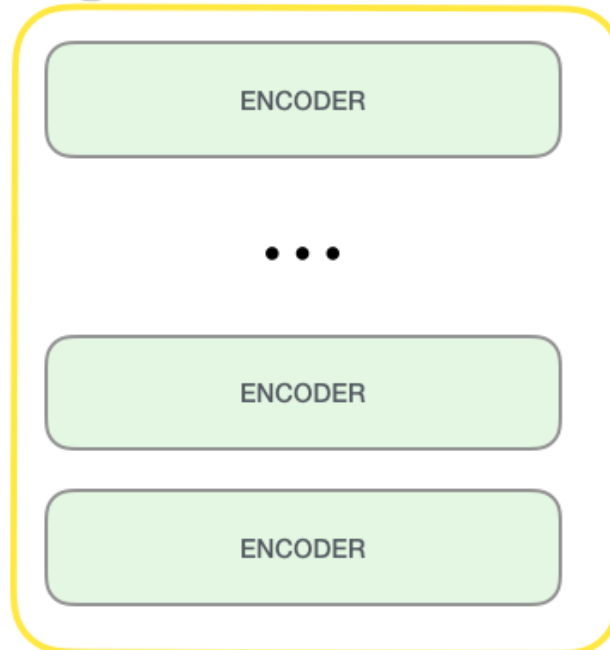
DECODER ONLY VS ENCODER-DECODER



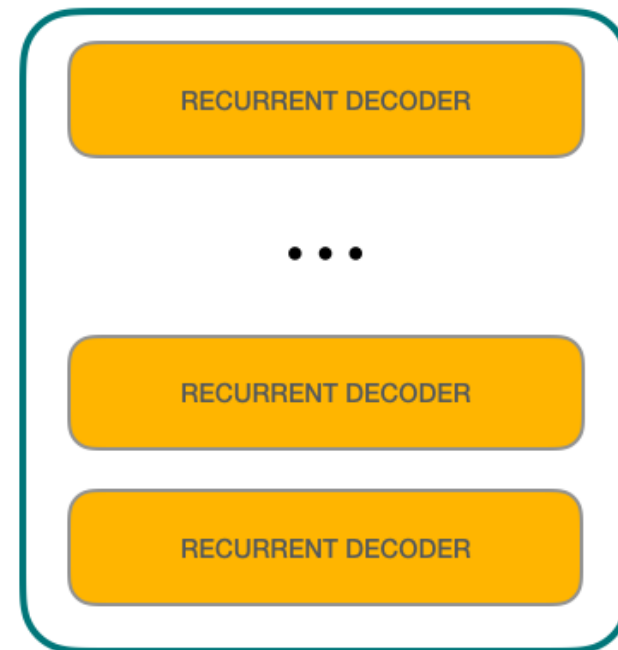
GPT-2



BERT



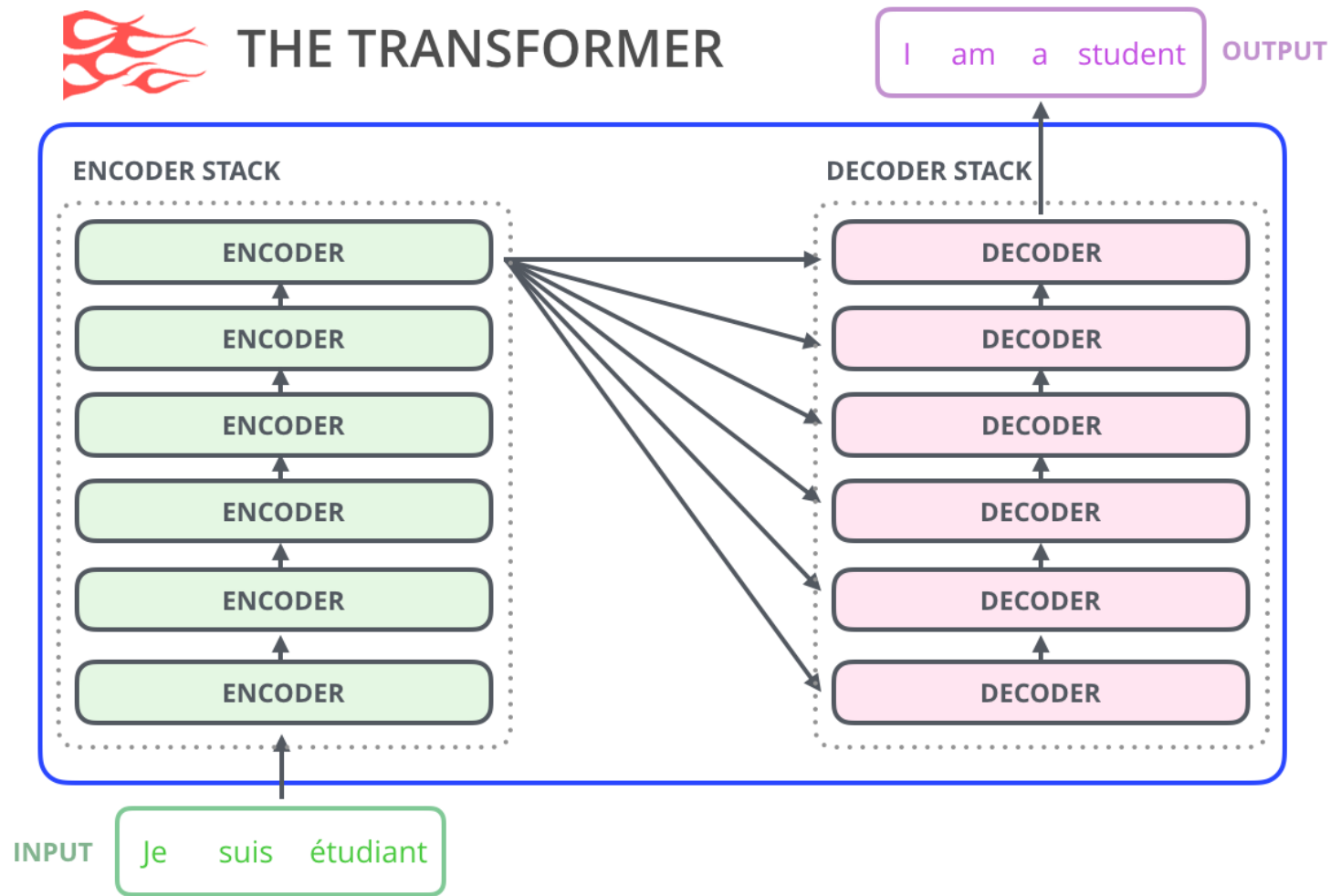
TRANSFORMER XL



DECODER ONLY VS ENCODER-DECODER

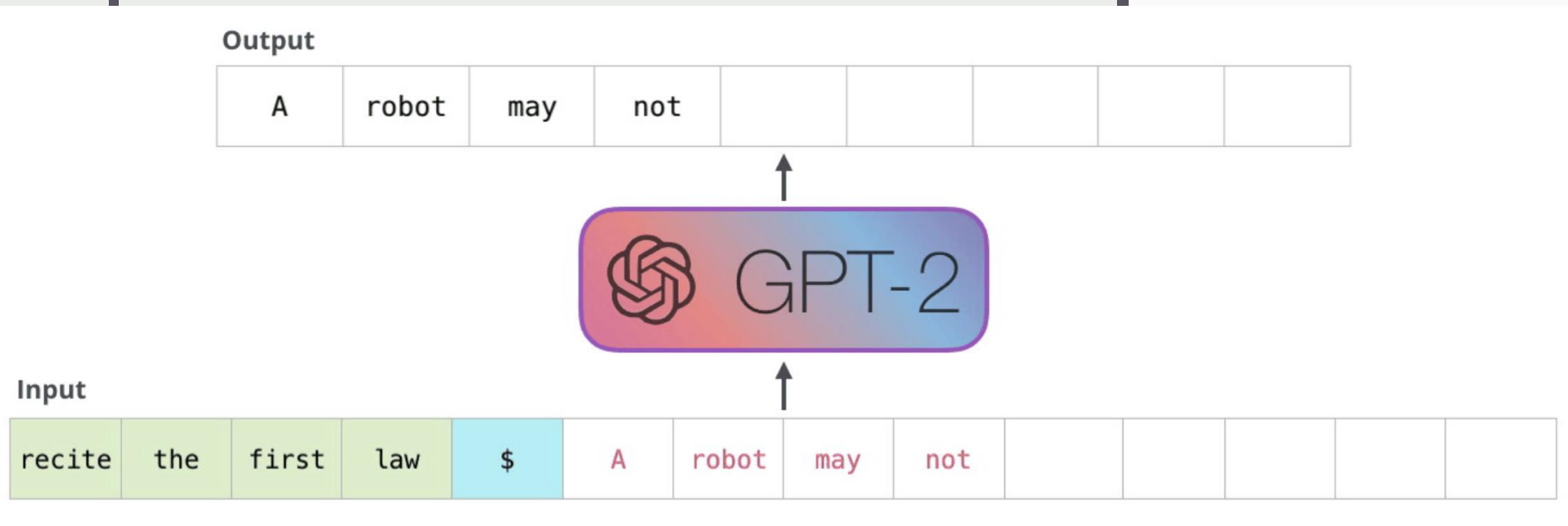
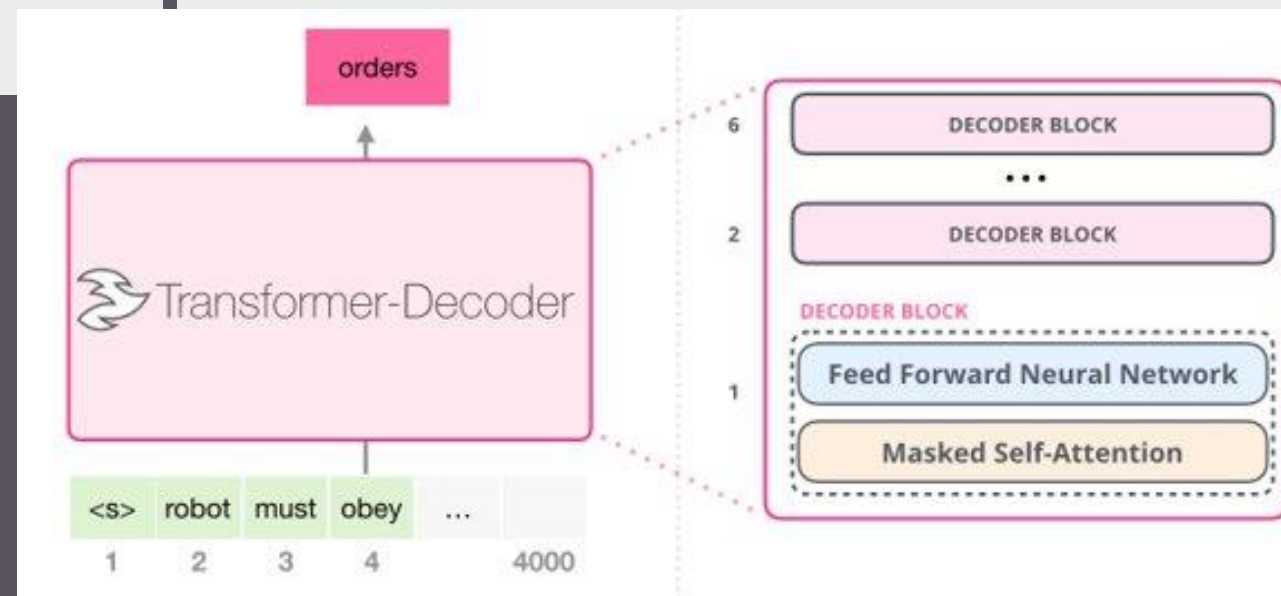


THE TRANSFORMER



ENCODER-DECODER (TRANSFORMER)

DECODER ONLY





SELECTING WHICH
WORDS TO GENERATE

Greedy Search

- Choose most probable word (or n-gram) given position

Greedy Search

- Choose most probable word (or n-gram) given position

Disadvantages?

Greedy Search

- Choose most probable word (or n-gram) given position

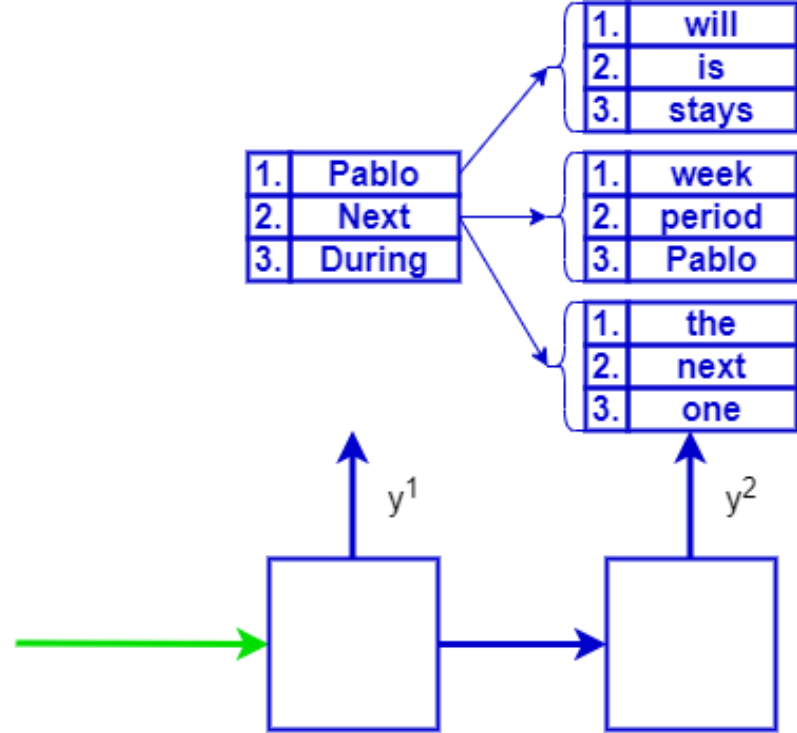
Disadvantages?

- Just because it is most probable, doesn't mean it is the "best"
- Always generate the same text from the same starting text
 - Boring conversations
 - No creativity
 - Etc.

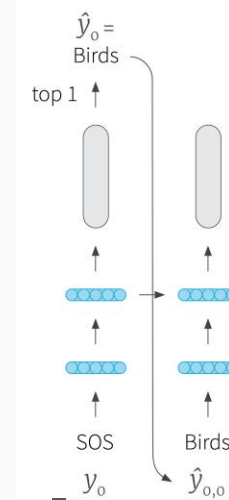
Beam Search

<https://towardsdatascience.com/foundations-of-nlp-explained-visually-beam-search-how-it-works-1586b9849a24>

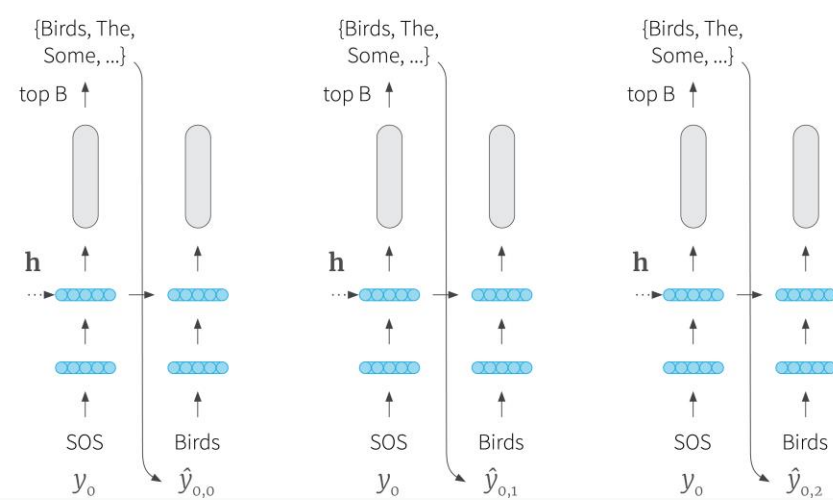
- With Greedy Search, we took just the single best word at each position. In contrast, Beam Search expands this and takes the best 'N' words.
- With Greedy Search, we considered each position in isolation. Once we had identified the best word for that position, we did not examine what came before it (ie. in the previous position), or after it. In contrast, Beam Search picks the 'N' best *sequences* so far and considers the probabilities of the combination of all of the preceding words along with the word in the current position



a) Greedy Search

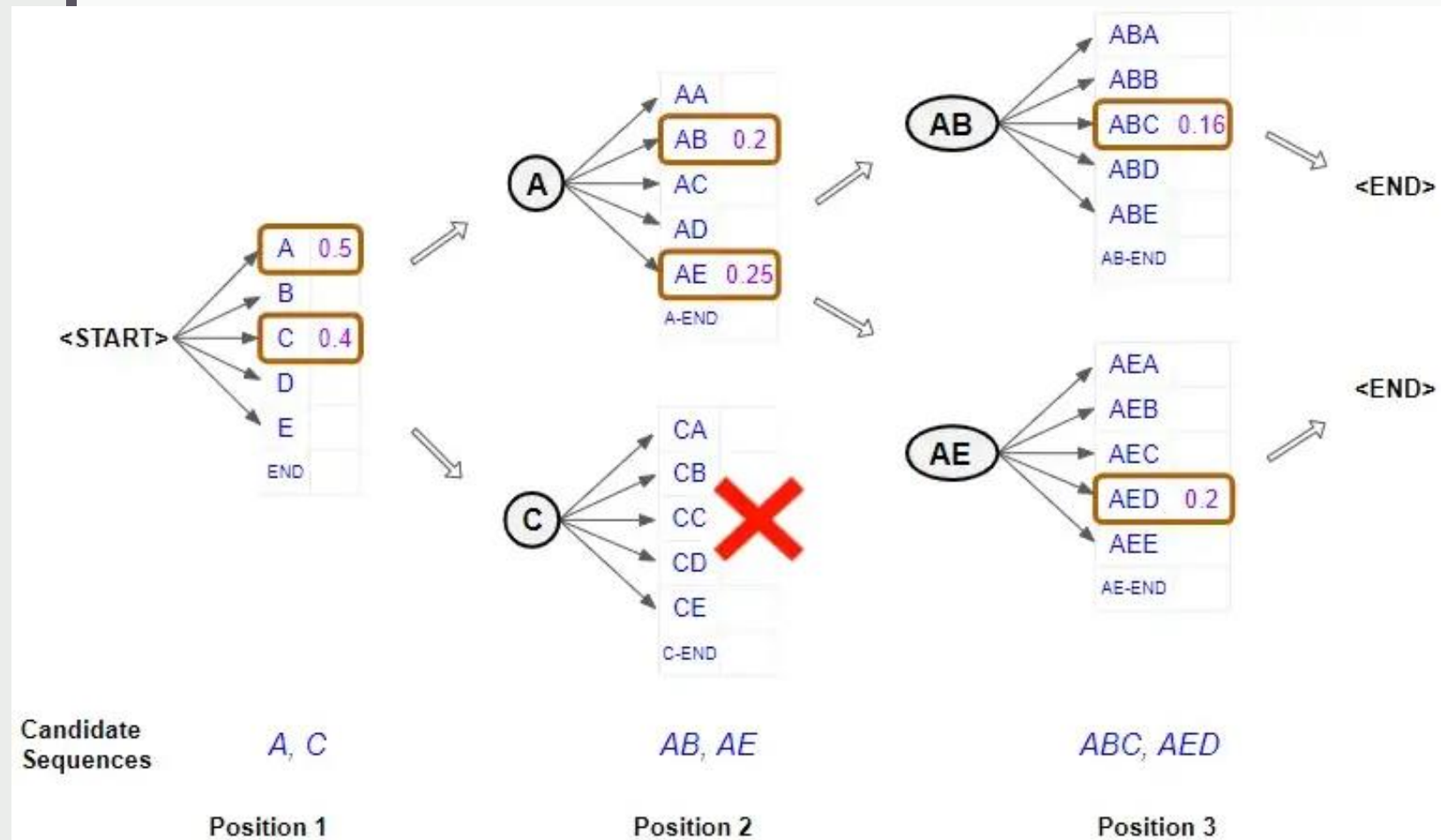


b) Beam Search



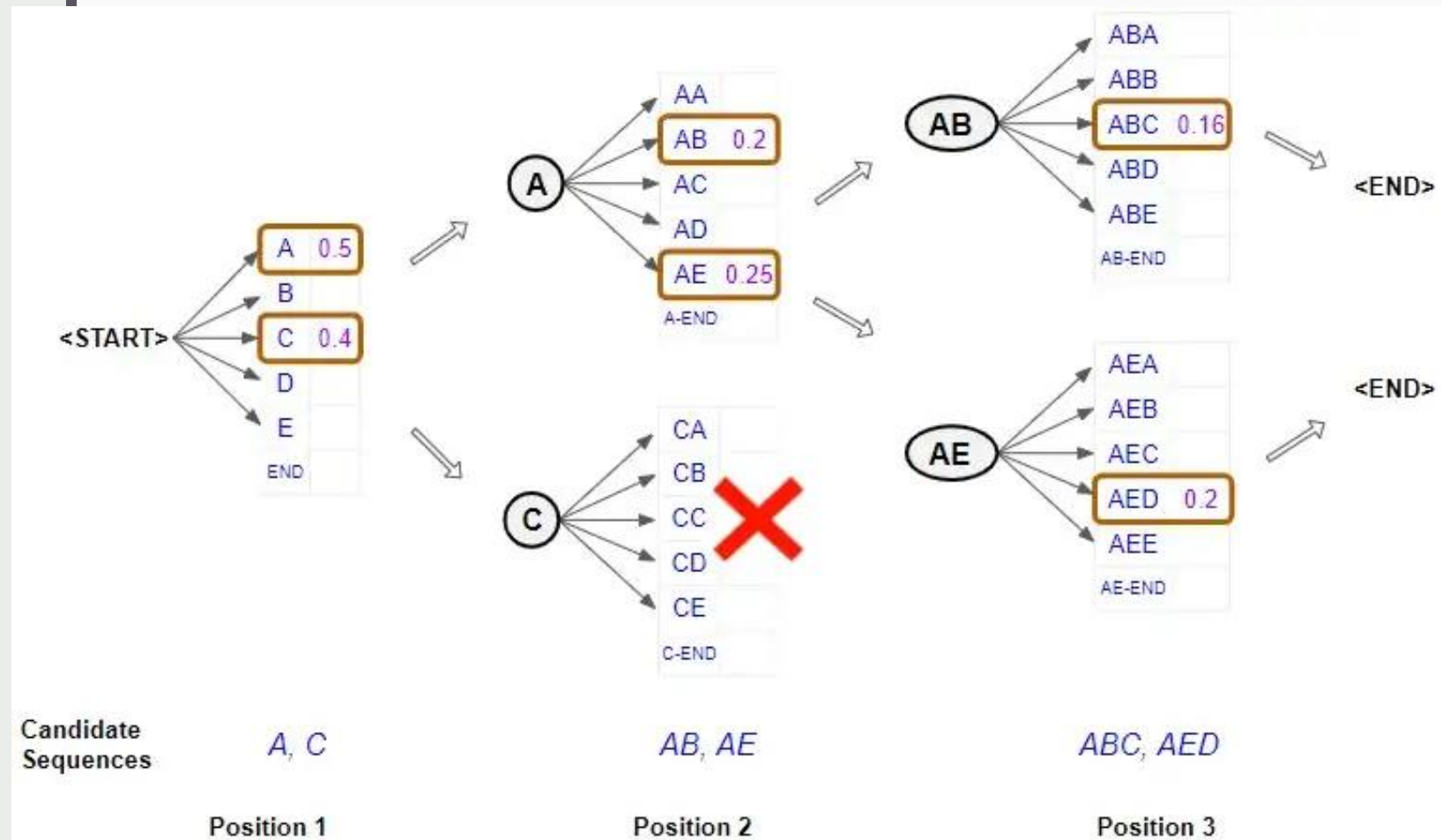
BEAM SEARCH

Beam Search



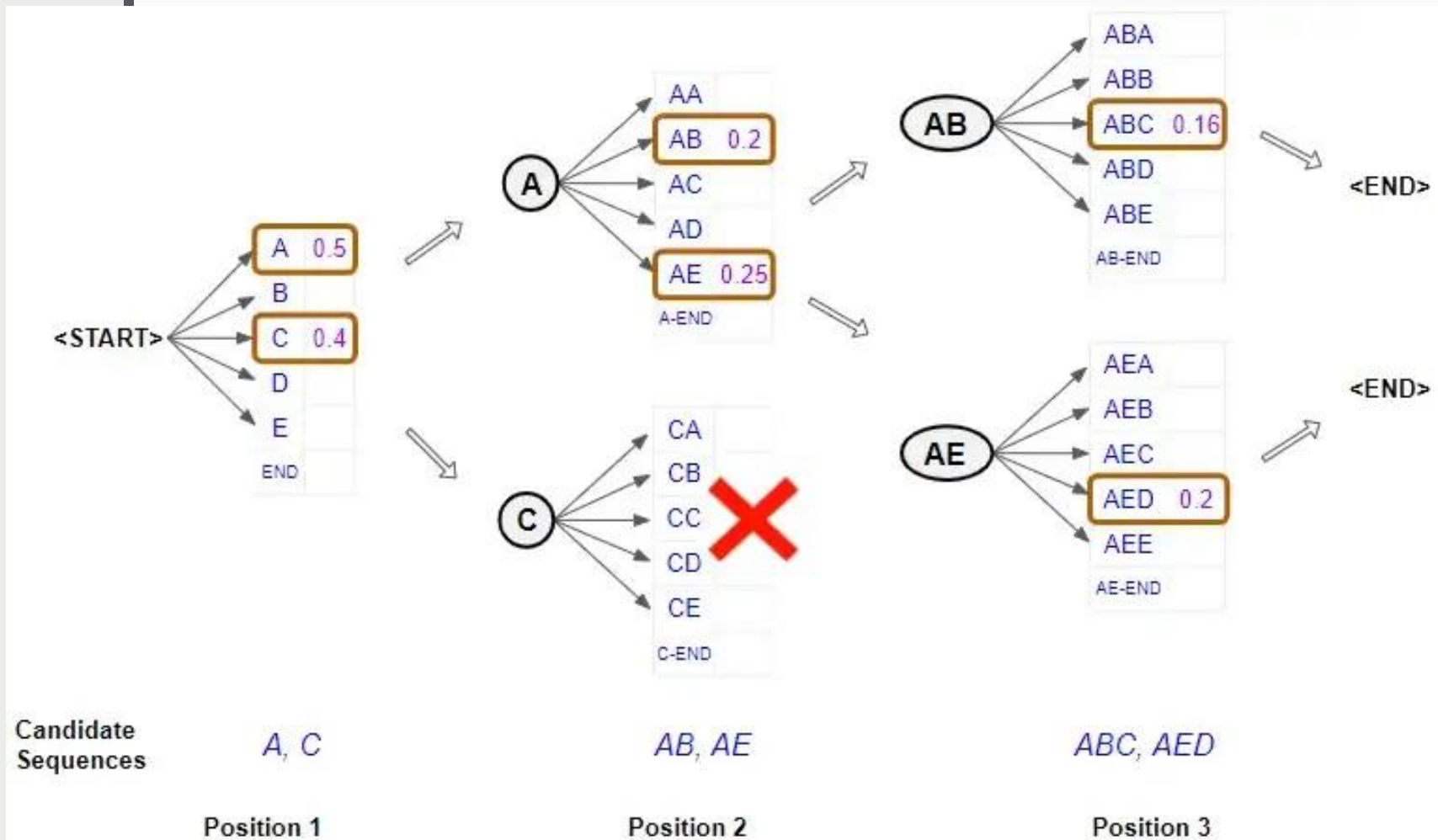
- Consider the output of the model at the first position. It starts with the "`<START>`" token and obtains probabilities for each word. It now selects the *two* best characters in that position. eg. "A" and "C".

Beam Search



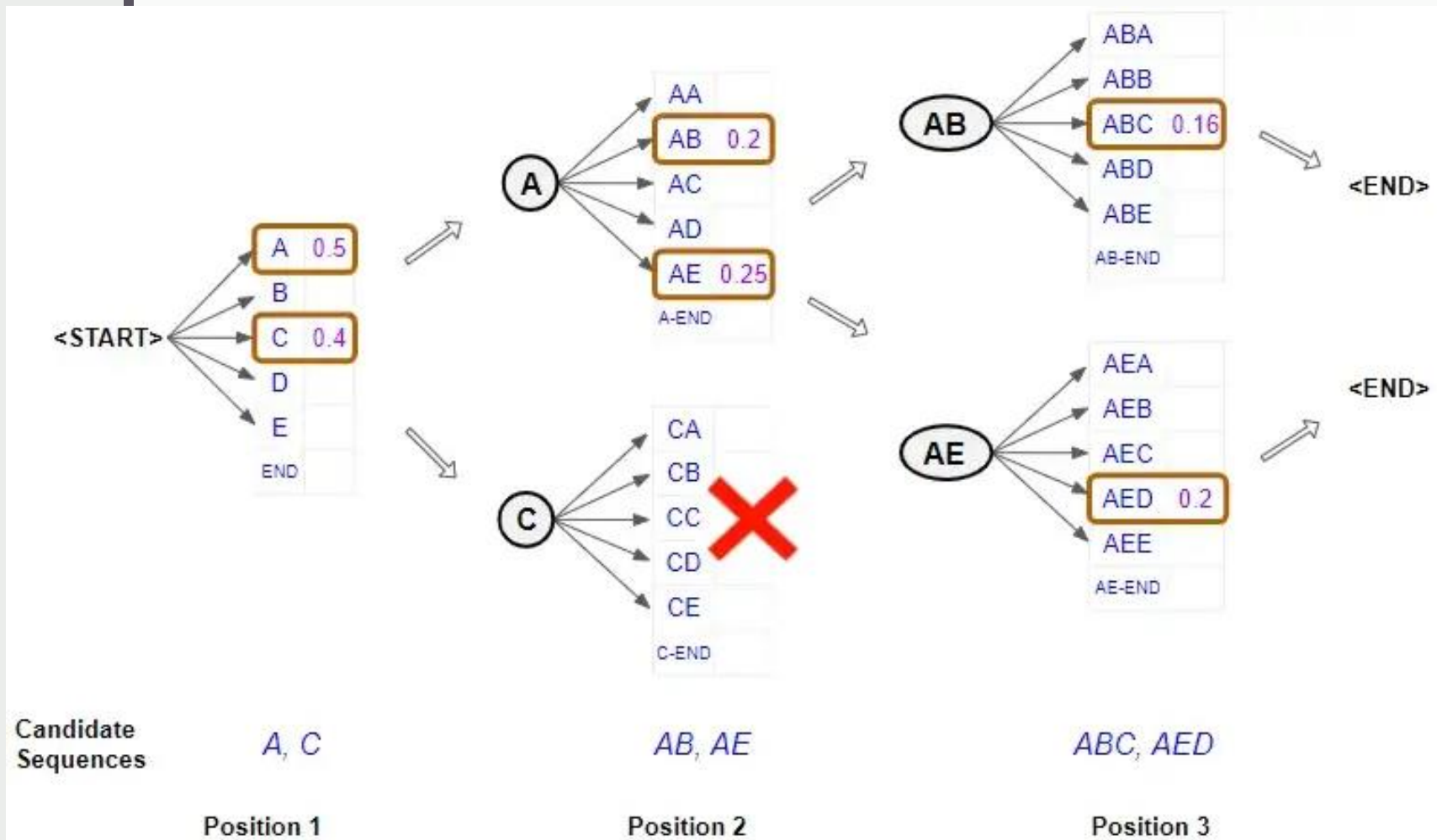
- When it comes to the second position, it re-runs the model *twice* to generate probabilities by fixing the possible characters in the first position. In other words, it constrains the characters in the first position to be either an "A" or a "C" and generates two branches with *two sets* of probabilities. The branch with the first set of probabilities corresponds to having "A" in position 1, and the branch with the second set corresponds to having "C" in position 1.

Beam Search



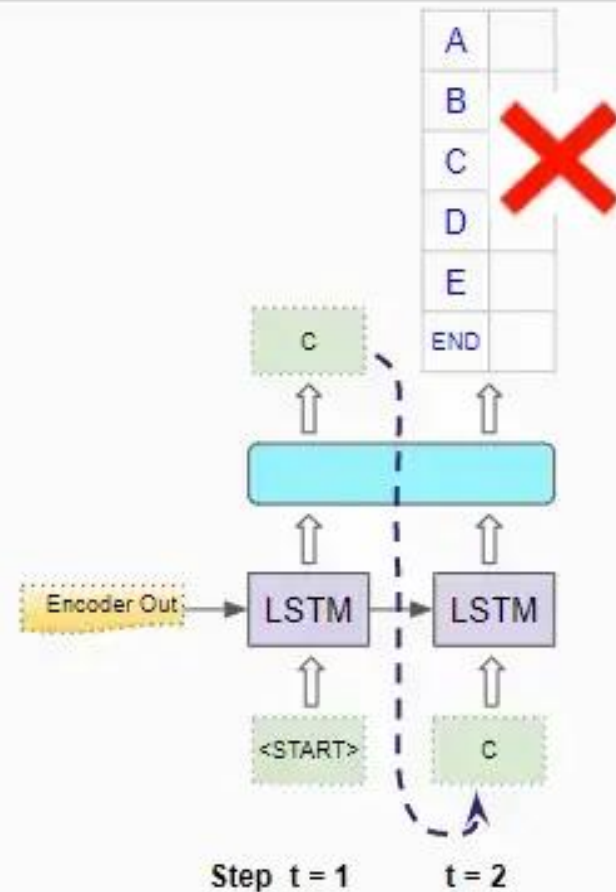
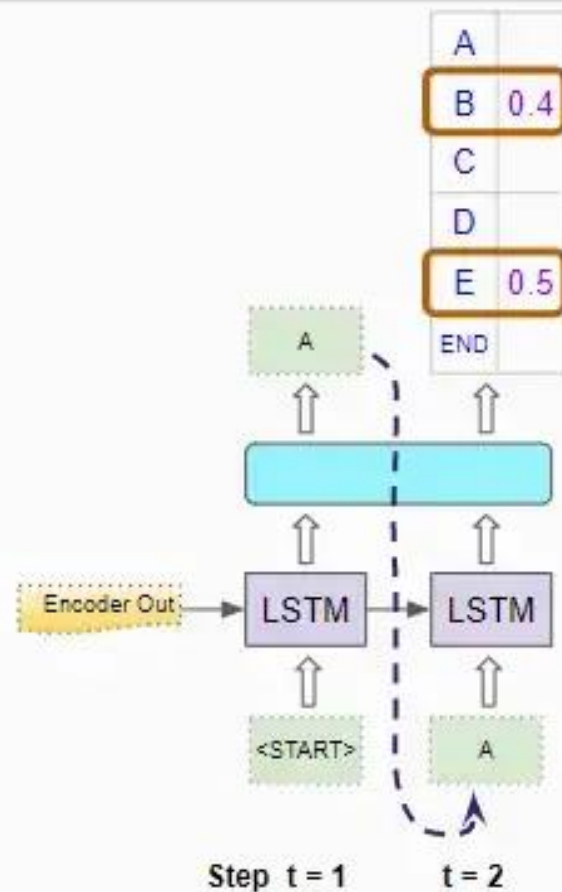
- When it comes to the third position, it repeats the process. It re-runs the model twice by constraining the first two positions to be either "AB" or "AE" and again generates two sets of probabilities.
- Once more, it picks the overall two best character triplets based on the combined probability of the first three characters from both sets of probabilities. Therefore we now have the two best combinations of characters for the first three positions. eg. "ABC" and "AED".

Beam Search

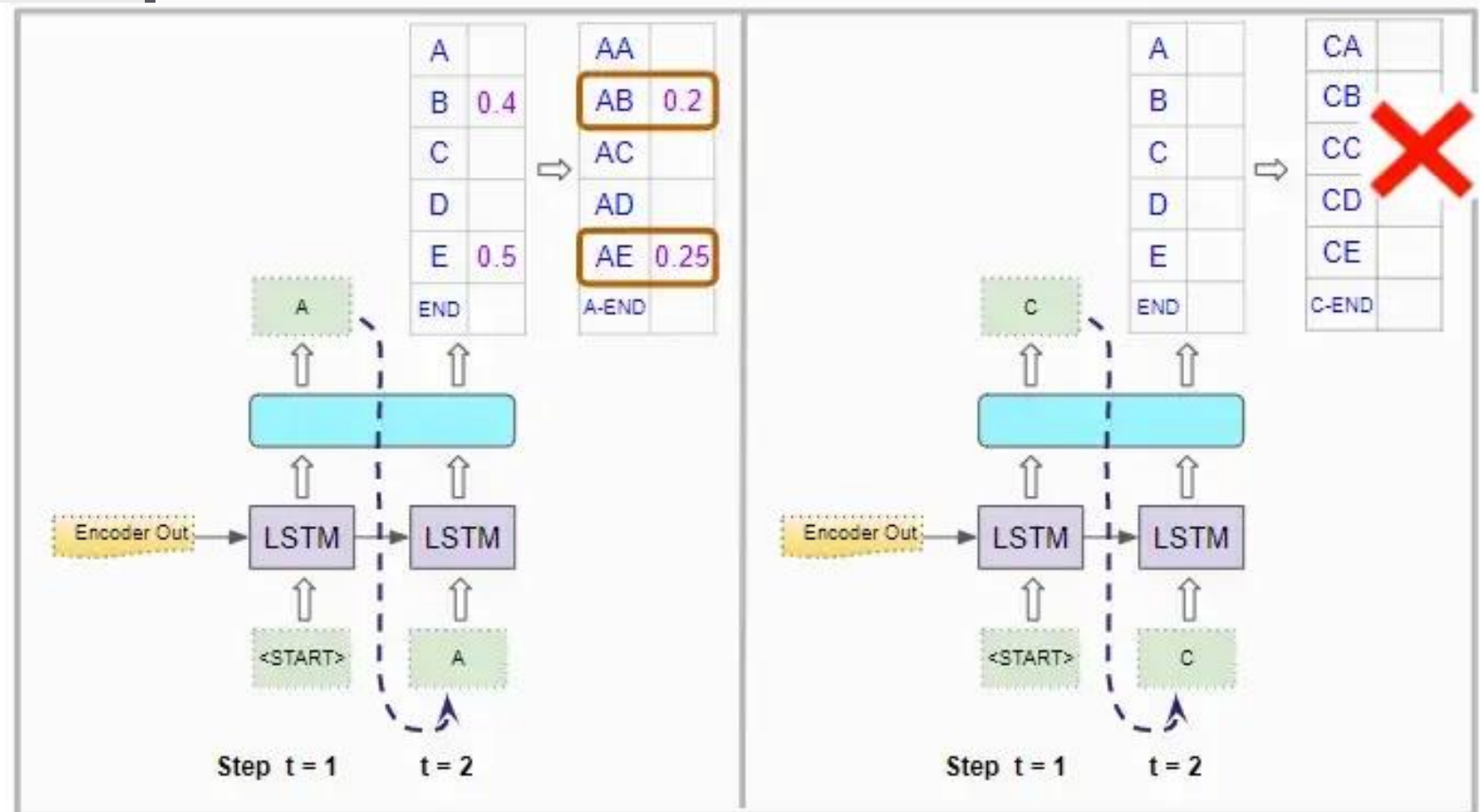


- It continues doing this till it picks an "<END>" token as the best character for some position, which then concludes that branch of the sequence.

Beam Search – Into the model

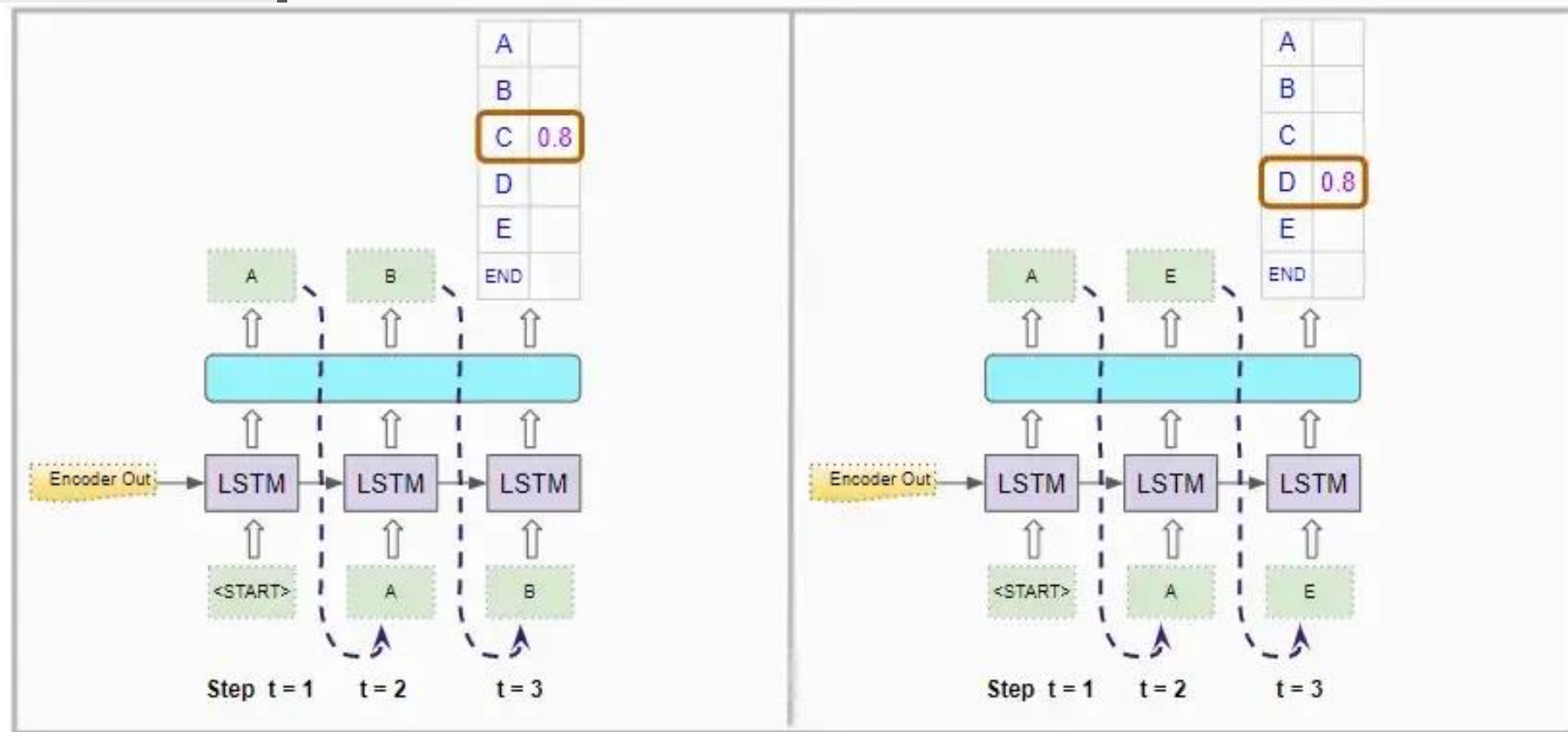


Beam Search – Into the model

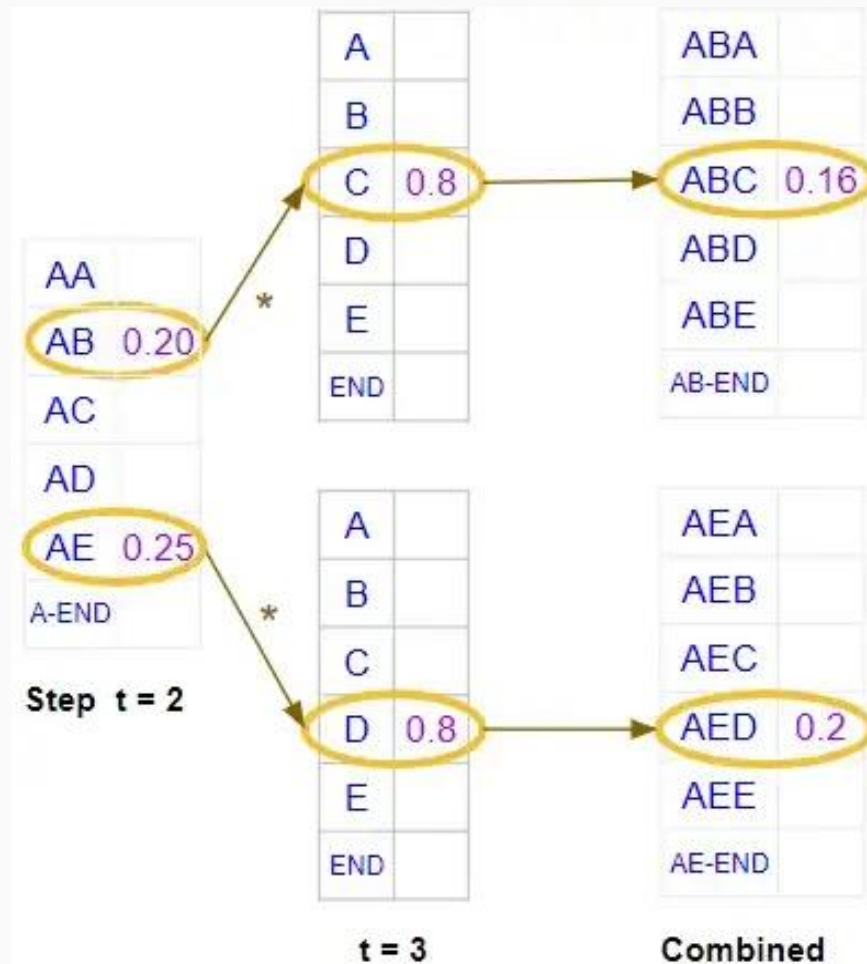


Position 2 : *AB*, *AE*

Beam Search – Into the model



Beam Search – Into the model



$$\begin{aligned}\text{Prob (ABC)} &= \text{Prob (AB)} * \text{Prob (C | AB)} \\ &= 0.2 * 0.8 = 0.16\end{aligned}$$

$$\begin{aligned}\text{Prob (AED)} &= \text{Prob (AE)} * \text{Prob (D | AE)} \\ &= 0.25 * 0.8 = 0.2\end{aligned}$$

Beam Search

Disadvantages?

Beam Search

Disadvantages?

- Still no variation in responses
- Beams can be very similar to each other, lack of diversity

"While beam search is superior to greedy search, it often produces sentences that have the same or a very similar start. Decoding is done in a left-to-right fashion and the probability weights are often concentrated at the beginning; even if search is performed over B sentences, the weight of the first few words will mean that most of these sentences start with these words and there is little diversity."

([Source](#))

Diverse Beam Search

Beam is divided into G groups.

Regular beam search is performed in the first group to generate $B' = B/G$ sentences.

For the second group, at step t of decoding, the beam search criterion is augmented with a factor that penalizes token sequences that are similar to the first t words of the hypotheses in the first group.

For the third group, sequences that are similar to those in either of the first two groups are penalized, and so on

Diverse Beam Search

4a) Beam Search

A steam engine train travelling down train tracks.

A steam engine train travelling down tracks.

A steam engine train travelling through a forest.

A steam engine train travelling through a lush green forest.

A steam engine train travelling through a lush green countryside.

A train on a train track with a sky background.

4b) Diverse Beam Search

A steam engine train travelling down train tracks.

A steam engine train travelling through a forest.

An old steam engine train travelling down train tracks.

An old steam engine train travelling through a forest.

A black train is on the tracks in a wooded area.

A black train is on the tracks in a rural area.

Figure 4. Diversity in beam search. Probability contributions tend to be concentrated near the start of the sequence, so that many of the sentences produced have very similar words at the start. Diverse beam search aims to increase the diversity by explicitly penalizing very similar hypotheses. Adapted from Vijayakumar *et al.* (2018).

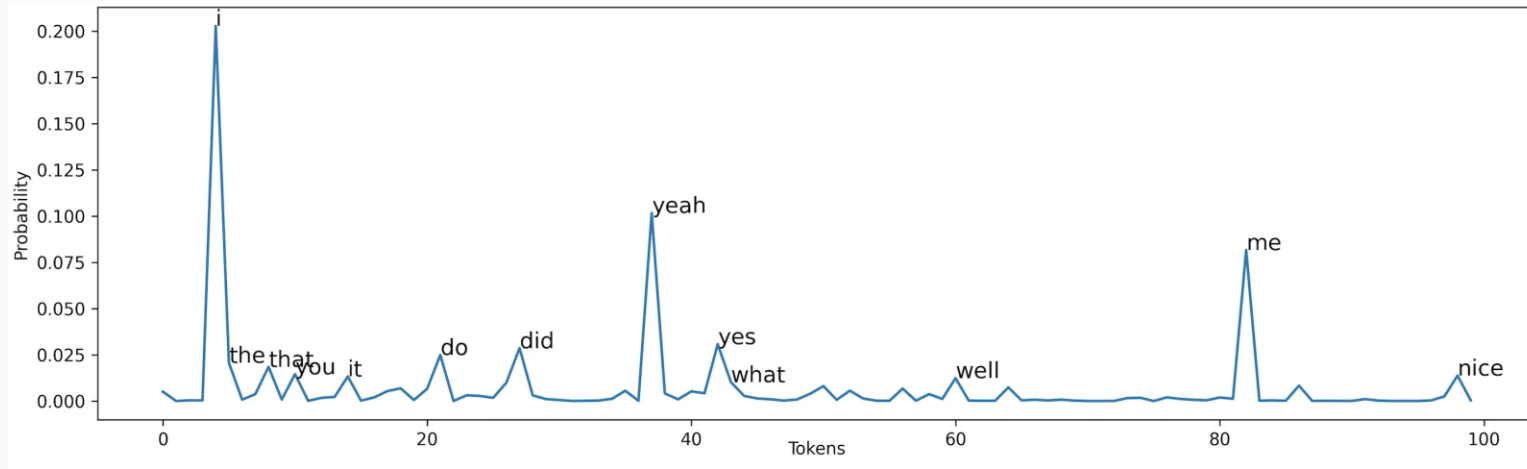
Diverse Beam Search

Disadvantages?

- More diverse, but still limited

RANDOM SAMPLING:

To avoid generating generic response, randomly choose response tokens based on probabilities

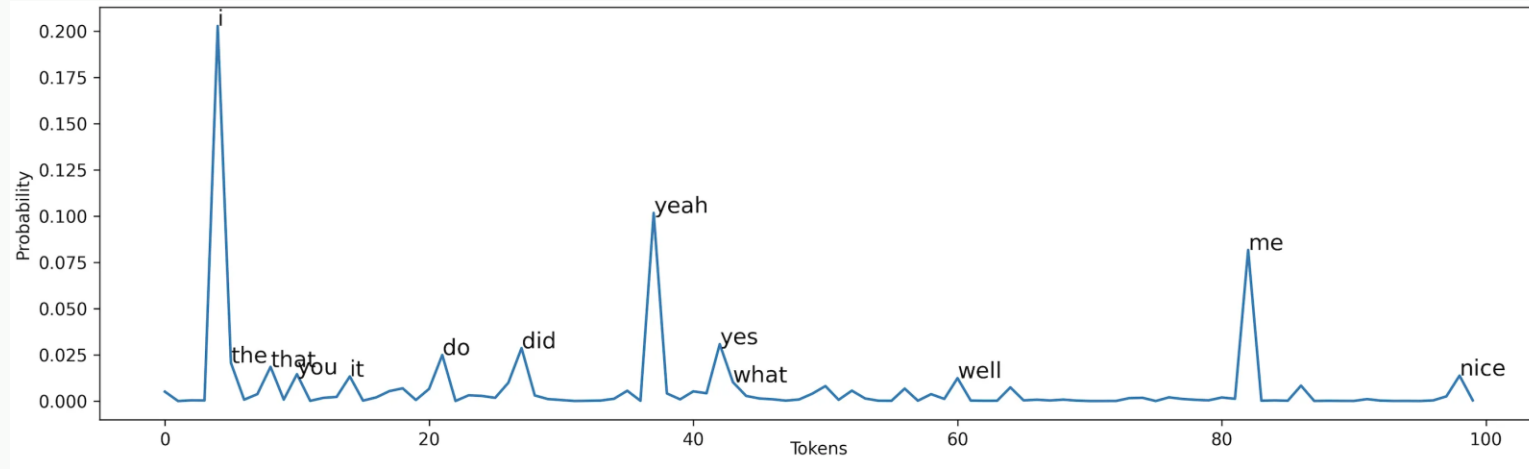


- Greedy search will result in "I" token
- Random sampling -> "I" has 20% chance to occur, but any can be chosen.

<https://towardsdatascience.com/decoding-strategies-that-you-need-to-know-for-response-generation-ba95ee0faadc>

RANDOM SAMPLING:

To avoid generating generic response, randomly choose response tokens based on probabilities



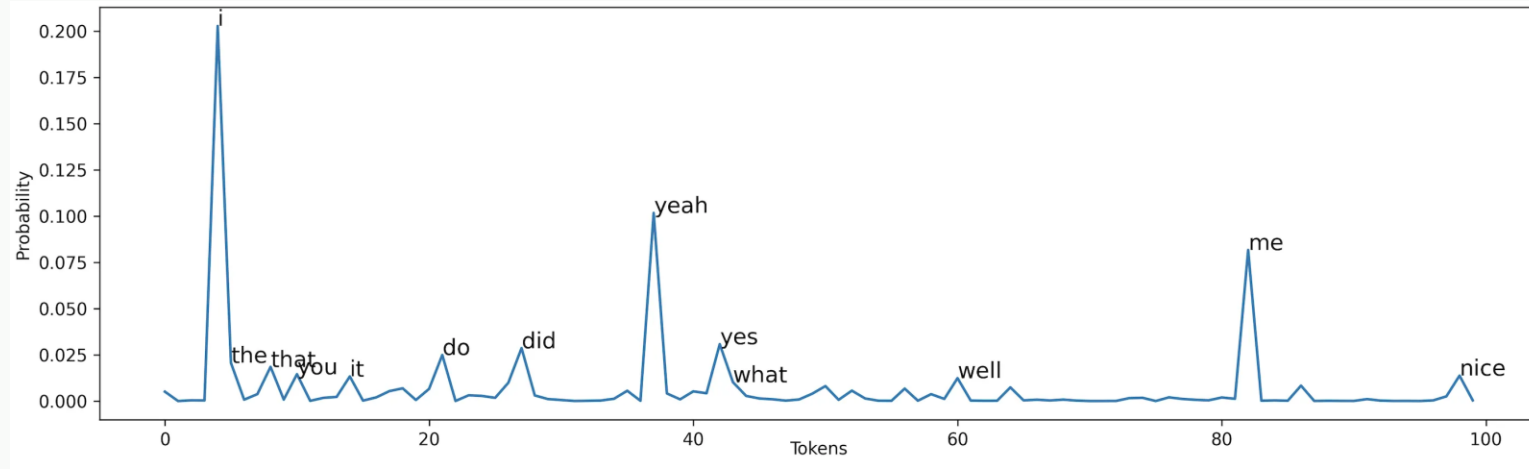
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Disadvantages?

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RANDOM SAMPLING:

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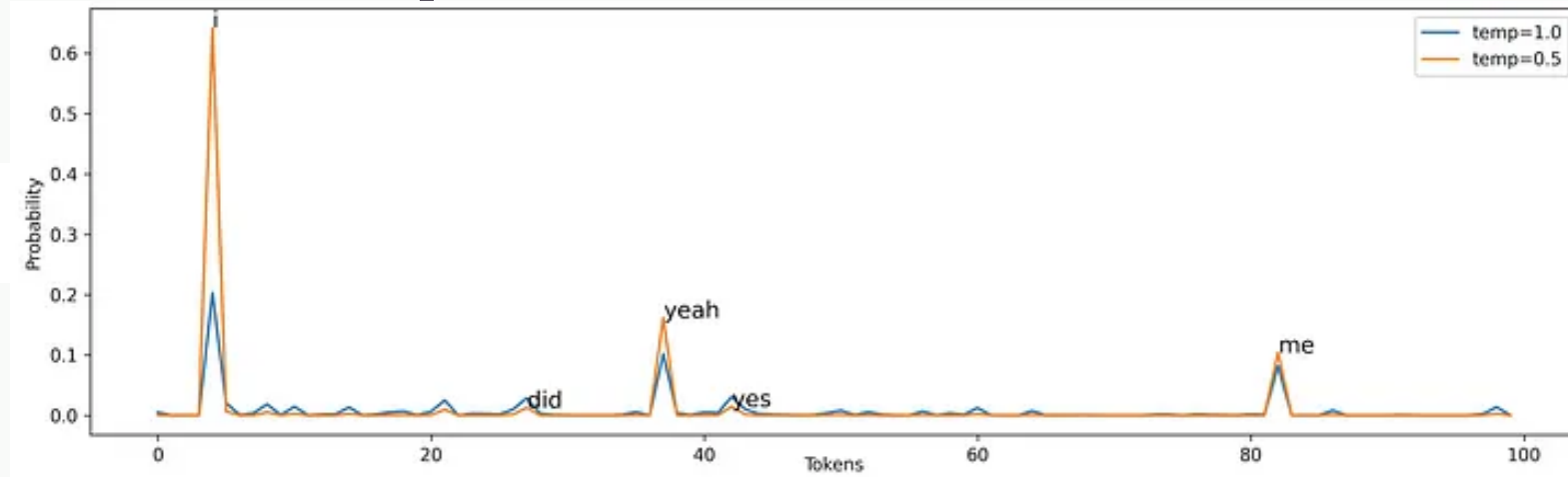
- Greedy search will result in "I" token
- Random sampling -> "I" has 20% chance to occur, but any can be chosen.

Disadvantages?

- Can result in poor word choices

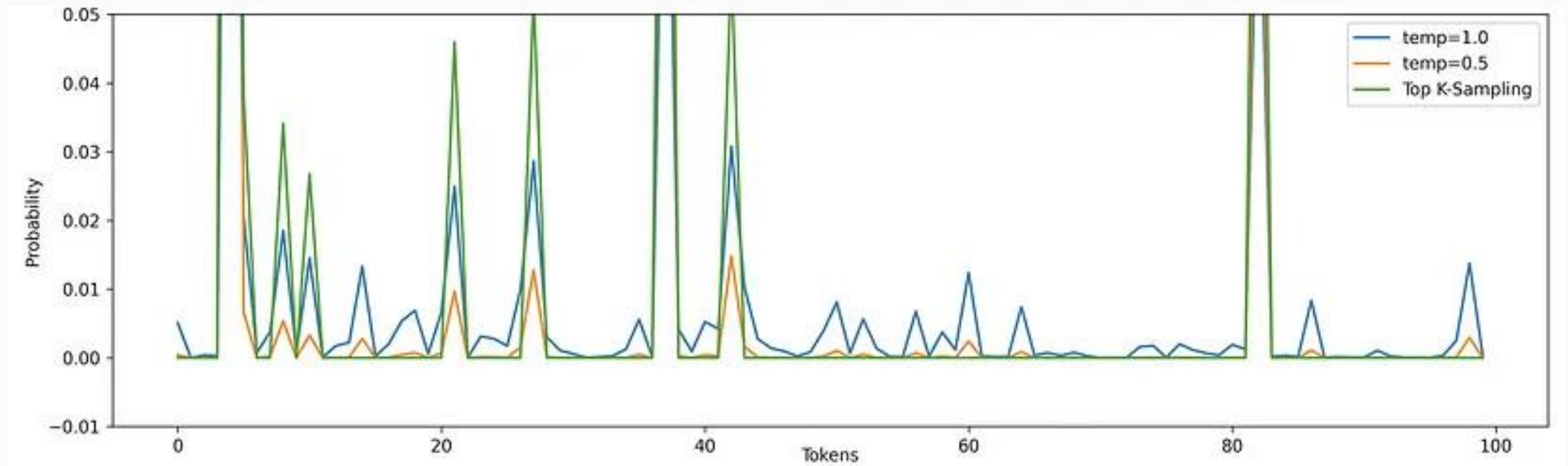
<https://towardsdatascience.com/decoding-strategies-that-you-need-to-know-for-response-generation-ba95ee0faadc>

$$P(x_i|x_{1:i-1}) = \frac{\exp(u_i/t)}{\sum_j \exp(u_j/t)}$$



Adding in Temperature

- Random Sampling can add in poor word choices
- We can add temperature to increase the probability of probable tokens while decreasing the probabilities of the less probable

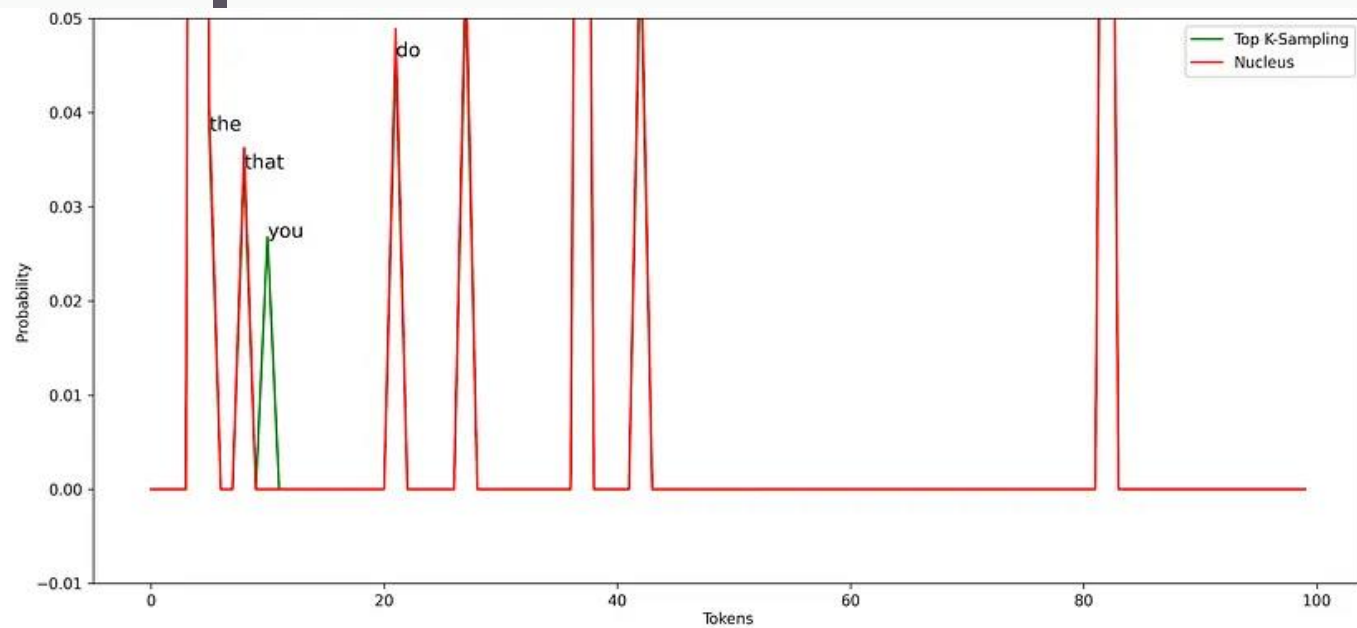


TOP – K SAMPLING

Rather than all tokens with probabilities to be drawn from, only allow choices from top -k.

- Could add temperature here as well

$$\sum_{x \in V(p)} P(x|x_{1:i-1}) \geq p.$$



Nucleus Sampling

- Rather than K, choose top-V which sum up to $\geq p$.
- Non V, set to 0, V rescaled to sum to 1. (100%)

NLG

Questions?